

CRITICAL THINKING PUZZLE BOOK

Five logic-driven puzzle sections designed to sharpen reasoning, pattern detection, and deduction.

HOW TO USE THIS BOOK

- 1) Choose a section.
- 2) Solve one puzzle per page.
- 3) Use pencil for grids.
- 4) Check solutions at the end.

Difficulty increases gradually within each section.

SECTIONS INSIDE

- 1) Mini Sudoku (6x6)
- 2) Nonograms (10x10)
- 3) Binary Puzzles (10x10)
- 4) Futoshiki (6x6)
- 5) Number Sequences

Solutions: last 10 pages (codes match each puzzle).

Section 1: Mini Sudoku (6x6)

S2-01

3	4	2	6		
6			3		
	3		2		6
	6		1		
4					2
5		3	4		

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-02

2	1		4		
		4			2
			5		
1	3		2	4	6
	2				3
	5	6		2	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-03

	4		1		5
6		1		2	
	6	4		3	2
3		5	4		
	3			4	
4					

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-04

2	6	4	5		1
	3		2		
1			4	5	
4					6
			3	1	
		2			5

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-05

	1	2	6	4	3
	6	3		5	
			2	6	
				1	
	5		4	2	
2				3	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-06

3	4	5		1	
	1			4	3
	6		2		4
4				6	
	2	1			6
		4			

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-07

	3	5	6		1
2	6			4	
			2		6
5					3
			1		
6		4		3	2

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-08

4					
5		1	4		
	4	6			
2	5	3	1	4	
	2		3		
	1	4			5

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-09

	4	2	6		1
		1	4	5	
	1			6	
		3			5
	5	6		1	
		4			6

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-10

	6				3
	5		4		
2		5	3	4	6
		1	5		4
	3	4	6	2	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-11

		3	2	4	
		2			5
		6	4		3
5			6	1	
3	4				
	6	5			4

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-12

5	4				1
6					
4		6	2	3	
		5			6
1	5			6	
	6	4			2

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-13

		5		2	6
6					
5	6		1		2
		3		4	
		6	2		
4	2			6	3

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-14

5	2				6
	6	3		5	
6	1	5	4		
			5		
		6		1	
	4	2			5

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-15

1	4	2			5
6					
			2	6	
		3	5		
	5	1	4		
	2	6	3	5	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-16

2		5	1	6	3
				2	
		3			
			3	5	6
4				3	
	1	2	6	4	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-17

	2	6	5		4
5					
		4			5
	5	2			
	6	5		3	
2		3		5	6

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-18

	2	1		5	
4	6			1	3
	1	4		3	
	5				6
			4		
	4	2		6	

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-19

		4	1	3	
5			2		
	6	1			3
				1	4
	3	6	4		1
1	4				

Fill 1-6 so each row/column contains every number exactly once.

Section 1: Mini Sudoku (6x6)

S2-20

5	3		4		6
	1	6		3	2
	4		2		1
	6				4
1		4	3		

Fill 1-6 so each row/column contains every number exactly once.

NG-01

Shade cells to match the run-length clues for every row and column.

NG-02

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-03

[illegible]

Shade cells to match the run-length clues for every row and column.

NG-04

NG-04

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-05

				1 1 1 1 2 2 1 1 3 1 2 1 1 2 2 4 1 1 1 1 2 2 2						
1	2									
3	1	1								
1	2	1								
1	1	1	1							
		1								
	2	1								
	1	2								
	1	2								
	3	1								
2	2	1	1							

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-06

[illegible]

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-07

[illegible]

Shade cells to match the run-length clues for every row and column.

NG-08

Shade cells to match the run-length clues for every row and column.

NG-09

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-10

				2 1 1 1 1 1 4 1 1 1 1 1 1 1 4 3 3 1 2 2 2 1 1 1						
4 1 1 3 1 2 2 1 1 2 1 1 2 1 2 1 1 2 2 1 2 1 1 1 1 3										

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-11

NG-12

Shade cells to match the run-length clues for every row and column.

NG-13

Shade cells to match the run-length clues for every row and column.

NG-14

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-15

										1
										1
			1							1
	2	2		1	1	3		1	3	2
	3	2	5	4	2	1	3	1	1	1
3	1									
1	4									
1	1	2								
2	1	1								
3	2									
2	1									
1	1	2	1							
2	3	1								
1	1	2								
2										

Shade cells to match the run-length clues for every row and column.

NG-16

Shade cells to match the run-length clues for every row and column.

NG-17

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-18

				1 2 1								
				1	4	1	1	1	1	1	1	1
				2	1	1	4	1	1	1	3	4
1 1 3 1 1 1 1 1 1 1												

Shade cells to match the run-length clues for every row and column.

NG-19

Shade cells to match the run-length clues for every row and column.

Section 2: Nonograms (10x10)

NG-20

				1 1 1 1 2 1 1 1 2 3 1 1 5 2 2 2 2 2 2 1 1 1 1 1 3 2 2 1 1										
1	1	4												
	1	2												
	3	3												
	1	3												
1	1	1	1											
	1	3												
1	3	1												
2	1	4												
2	1	1												
1	2	1												

Shade cells to match the run-length clues for every row and column.

Section 3: Binary Puzzles (10x10)

BP-01

1	0	0					0		0
	1	1			0		1	0	1
	1	1		0			1	1	1
	1			0		1			
1					0				
					1	1		1	0
			1		1				0
	0		1	0	0				
1	1		0	0		0	1		0
	0		0	1				1	1

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-02

1		1	1				0	1	
					1	0			0
1	1		0	1					1
	0	1	1	1		0			
		1				1	0	0	
		0		0		1		1	
1		0	1		0			1	0
0		0		1				0	1
	0					1	1		0
0	1			0				0	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-03

0	1		0			0			1
0		1		1		1		1	
1				0			1		0
		0		1	1	1		0	
			1	0	0	0		0	
0	1	0	1		0	1	1	1	
				0				1	0
	0		1			0			
1			0			0	0		
							1	0	0

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-04

1	0		0	1		1		1	
0	1							0	
	1		1				1		0
0					1	1		1	1
				0	1			0	
1			0		1				0
1					1	1		1	0
	1		0			0		1	1
1				0	0			0	1
0		1		0		1			0

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-05

1		1	1	0					
0	1		1	1	1	1	0		0
			1			0	1		
	1						1	1	
0	1	0	0	1	0		1		1
		1	1		1				
1				1		0			
1	0			0			1		
1	0		0			0	0		
1					0		1	1	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-06

1			1		0	0			1
		1			0	1		0	0
1	1						0		0
						0	1		
		1	1	1	0				1
		1	1	0	0			0	1
		0		0	1				
0		0			1		1		1
	0			1	1		1		0
	1		1			1		1	1

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-07

	0				0		0		
		1				0	1		0
			1	0	1		0	1	0
	1	1			1			0	
		0	0				1	1	0
	0	1	0			1	1		0
		0				0	0		1
				0			1	0	1
0		1		0					1
1	1	0	1		0				

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-08

	1	0		1					
1		1		1	0	1		0	
		1					0	1	1
0		0			1		1	1	0
		0		1		1			0
1	0	1		0		0			
	0					0	1		1
0	1		1				1		
		1	0	1				0	1
1	1				1			0	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-09

	1		0		1			0	
0		1			1		1		1
0				0				1	1
1			0			1		0	
0	1	0			0		0		0
1		1		0	0	1		0	1
	0	0	1		1				
			1					0	0
0			0			0	1		
		0	0	1	1				

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-10

0	1		1	1			0		0
1				1	0		0		0
	0	0	0	0					
	1					1			
1	1	0			0				
0	0	1	1			0		0	
1						0	0		
0	0			0		0			1
	1	0	0					1	
0			1	1	0	0	1		

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-11

	0	1			1	0		1	
0		1					1		
	1			1	0		0	0	0
		0		0			1	0	
1	0	0	1	0		0	0		
0		0							
			1		0	0			
0	1			0			1	0	
	1	1	1		0			1	
1		0				1	1		1

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-12

1			1			0		0	
1				1		0			
0				1	1		1	1	0
						0	0	0	
	0						1	0	0
0	1	0						1	1
		1	0	0		1	0		
			0	0	0				0
1	0	1	0	1		1		0	1
			1	1					1

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-13

1		1	1	0		0		0	
		0				0			1
	1	1					0		1
0		0						1	0
				1	0		1		0
1	0			1				1	
	1	0		1		1			1
1	0			0			1		
	0				1	1		0	
	0		0	1	0	1		1	0

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-14

		1			0			0	
	1			0		0		0	0
				0		1	1	0	1
	1	0	0			1			
1		0	1		1	0			
					1			0	
0		1	1	1	0			1	1
		0			1			1	
0	0	1	1			0			1
			0			0	0	1	1

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-15

0	1		0	1			1		0
	0				1	0			0
0		1				1		1	0
	1	1					0	0	
	0	0	1		1				
	1		1	0		1			1
		1	1	0	0		0	0	1
0				1					0
		0	0						1
				0	0	1		0	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-16

	0			1			1	0	
		1	0	0		1	1	0	
0		1	1		0				
		1	0						
1	1	0				1	0	1	0
		0	0					0	
1		0	1				1	1	0
0	1				1		0	1	
	1	0	0				0		1
			1	1				0	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-17

						0	1		0
0				1	0			1	
0					0			0	1
	1	0	1	0		1	0	0	0
1						0	1	0	
1	1				1	0	0	1	
0	0		1						1
				0				1	
0	0	0		1		0	1		1
1					0	1			

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-18

	1	0	1		1	1	0		0
		1						1	0
	1	0		0			1		
		0	0					0	
		1			0		1	1	1
1			0	0			1	1	0
1	0		0				0	0	
0		1	0				1	0	
1	1	0		0					0
			1					0	

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-19

	0				1		1	1	
	0	0		1					
0		1	1	0	1	0	0		
		0			1	0		1	
	1	0			0	0			
1	1	0	0		1				0
	0	1	0					1	
0	0	1							
1	0		0		0	1	0		
		1	0				1		0

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 3: Binary Puzzles (10x10)

BP-20

1	1		1	0					
	0	1				1	1	0	0
1	0		1				1		0
0	0	1		1		0	0		
0		1		0			0		
1			0				1		0
			0		0		1	1	
		1		1			0		
					1	1	1	0	1
	1	1			1	0			

Rules: 1) No more than two identical numbers adjacent.
 2) Each row/column has equal 0s and 1s.
 3) No two rows or columns are identical.

Section 4: Futoshiki (6x6)

FT-01

6	>	1		<	4			
v					^			
	<		6	>	5			
^		^	v					
			4				>	1
v			^	v		^		
1		3	<		2			4
	>			<			<	
				v		^		
			3		1			6

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-02

				6	
			^	v	
		3		>	4
^		^			<
	5			>	1
^	v	^			2
	4	<	1		
v					^
	<	6	>	3	<
^		^			v
	1	<	2	5	4

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-03

1			2	6	3
	v	v			v
4	>	3	<		>
3			4	>	
			v	v	v
6		<	>		<
	v	^			
	>	<	5		4
				^	
	2		>	1	<

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-04

	<			>			
	v		^		v		
1	3	2	<				
	^	v		v			
		1	4		3		
		^		^			
	2	<	4	>	<		
	1	<		>	4	>	2
	^						
	6	5					1

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-05

		>	1	3		2
			^		v	^
	<			5	>	<
^		v		v		^
		1			3	
^		^		v	^	
5	<		>		<	1
^		v				
	>		>	2	1	<
			^			3
v						
3	>		<	5	<	6
					>	
						<

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-06

	2		>		3
v					
	>		>		4
2			5	>	3
				>	1
5		>	1	<	3
				>	2
					^
	3	>		<	
			v		
	<	5	3	<	

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-07

	1	<		>	5
					v
>			<	5	>
	^				^
<	5		3	>	1
	v	^		^	<
1	3		>	<	6
				6	>
				v	v
2	<			<	
		^			v
5		<		3	2

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-08

	<				> 4
		v			
	>	4	<	3	
		^			
1			2	5	3
^		v		v	^
	6		<		
^		^	^	^	
5			6		2
	^				
		> 2	3		< 6

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-09

	5		>	1	6	
^	v					
		>	1	<	3	<
	<	4		>	3	<
						2
					v	
3	6				<	
		v		^	v	
4					>	2
					v	
			<	5	>	4

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-10

5		<		>	3		>	
	^							^
				>	1		<	6
	v		^			v		
	1			<		>		
^								v
	>	4		>	2	>		3
		^						
3			1			>	2	4
		v						
1	<					3		

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-11

2	1			<	6
	^		v		
1	<		>	<	
4		>	1	<	6
	v				>
6	4		5	>	<
	^	^			v
	<	6		2	
	v				
3					

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-12

	>		>	1	<
		^			^
2	<	5	1	<	6
		v			^
	>	2	3	<	1
			^	v	^
	>	1	5		4
v				^	
	<				>
					2
					^
6			<	4	<

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-13

		6		2	
		v		^	v
	<			>	
					^
5	>	2		3	<
v		^			6
1	4	3		5	
					^
			<		
	^	^			
	6	5	2		<
					4

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-14

5			3 ^	<	<	6
					>	2
v	v	^				
	< 4 >					
	^					v
		4		2	<	3
^	v		v	^		v
	2	6	5		>	
6	3		> 2 >			

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-15

	>	3	4		>
^		^	^		
4	>		5	1	2
		v			v
3	<		6	>	2
^		v			
		1	<	2	
	v	^		^	
6			>	<	4
v			^		
1				<	

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-16

				1	<		<
	v		v			^	
	2						1
						v	
5	>		<	3		2	
v							
			2	<		>	5
	v						
6			>	1	<		4
							5
						v	
	<			5	>	3	

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-17

6	>	2	>	1	<		3	4
							^	^
2		1						
^		^			v			v
3			<			<		
		v			^		v	
5	>				>		1	<
v		^			^		^	v
4		5	>		<	6		
v					v			^
					<		5	>

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-18

	>				
		v			^
5		>	1	2	
		^	^		v
1		2	4	<	5
			v	v	
6				3	
		v			
4	>	1		>	<
v				v	5
2		4			

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-19

5	2				>	1
1		5	2		>	
∧		∧				
		6		3		2
	<	4	>		>	2
∨			∨			
	<			<		
			∧			
6		>	<	3	>	1
					<	

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 4: Futoshiki (6x6)

FT-20

5	6	>	4	2	>	3
				^		
		<		3	2	
				^		
					>	
v				v		
1	2	<	3	<		>
^			v	^		
2				<		<
	^			v		v
4	<			1	>	2

Fill 1-6 with no repeats in any row/column. Respect all inequality signs.

Section 5: Number Sequences

N01

Find the next number in the sequence:

3 , 12 , 48 , 192 , 768

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N02

Find the next number in the sequence:

1 , 2 , 3 , 5 , 8

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N03

Find the next number in the sequence:

12 , 16 , 20 , 24 , 28

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N04

Find the next number in the sequence:

1 , 3 , 7 , 13 , 21

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N05

Find the next number in the sequence:

8 , 64 , 512 , 4096 , 32768

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N06

Find the next number in the sequence:

2 , 5 , 8 , 11 , 14

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N07

Find the next number in the sequence:

4 , 12 , 36 , 108 , 324

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N08

Find the next number in the sequence:

4 , 2 , 6 , 8 , 14

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N09

Find the next number in the sequence:

7 , 11 , 15 , 19 , 23

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N10

Find the next number in the sequence:

3 , 5 , 9 , 15 , 23

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N11

Find the next number in the sequence:

7 , 49 , 343 , 2401 , 16807

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N12

Find the next number in the sequence:

4 , 7 , 10 , 13 , 16

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N13

Find the next number in the sequence:

1 , 2 , 4 , 8 , 16

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N14

Find the next number in the sequence:

2 , 1 , 3 , 4 , 7

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N15

Find the next number in the sequence:

23 , 27 , 31 , 35 , 39

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N16

Find the next number in the sequence:

2 , 4 , 8 , 14 , 22

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N17

Find the next number in the sequence:

2 , 4 , 8 , 16 , 32

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N18

Find the next number in the sequence:

7 , 13 , 19 , 25 , 31

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N19

Find the next number in the sequence:

1 , 4 , 16 , 64 , 256

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

Section 5: Number Sequences

N20

Find the next number in the sequence:

2 , 5 , 7 , 12 , 19

Answer: _____

Tip: look for arithmetic/progression, Fibonacci-like rules, differences, or powers.

SOLUTIONS

Solutions Page 1/10

S2-01

3	4	2	6	1	5
6	5	1	3	2	4
1	3	5	2	4	6
2	6	4	1	5	3
4	1	6	5	3	2
5	2	3	4	6	1

S2-02

2	1	3	4	6	5
5	6	4	3	1	2
6	4	2	5	3	1
1	3	5	2	4	6
4	2	1	6	5	3
3	5	6	1	2	4

S2-03

2	4	3	1	6	5
6	5	1	3	2	4
1	6	4	5	3	2
3	2	5	4	1	6
5	3	6	2	4	1
4	1	2	6	5	3

S2-04

2	6	4	5	3	1
5	3	1	2	6	4
1	2	6	4	5	3
4	5	3	1	2	6
6	4	5	3	1	2
3	1	2	6	4	5

S2-05

5	1	2	6	4	3
4	6	3	1	5	2
1	3	4	2	6	5
6	2	5	3	1	4
3	5	6	4	2	1
2	4	1	5	3	6

S2-06

3	4	5	6	1	2
2	1	6	5	4	3
1	6	3	2	5	4
4	5	2	3	6	1
5	2	1	4	3	6
6	3	4	1	2	5

S2-07

4	3	5	6	2	1
2	6	1	3	4	5
1	4	3	2	5	6
5	2	6	4	1	3
3	5	2	1	6	4
6	1	4	5	3	2

S2-08

4	6	2	5	3	1
5	3	1	4	6	2
1	4	6	2	5	3
2	5	3	1	4	6
6	2	5	3	1	4
3	1	4	6	2	5

S2-09

5	4	2	6	3	1
3	6	1	4	5	2
4	1	5	2	6	3
6	2	3	1	4	5
2	5	6	3	1	4
1	3	4	5	2	6

S2-10

4	6	2	1	5	3
1	5	3	4	6	2
3	4	6	2	1	5
2	1	5	3	4	6
6	2	1	5	3	4
5	3	4	6	2	1

SOLUTIONS

Solutions Page 2/10

S2-10

6	5	3	2	4	1
4	1	2	3	6	5
1	2	6	4	5	3
5	3	4	6	1	2
3	4	1	5	2	6
2	6	5	1	3	4

S2-12

5	4	3	6	2	1
6	2	1	5	4	3
4	1	6	2	3	5
2	3	5	4	1	6
1	5	2	3	6	4
3	6	4	1	5	2

S2-13

1	3	5	4	2	6
6	4	2	3	5	1
5	6	4	1	3	2
2	1	3	6	4	5
3	5	6	2	1	4
4	2	1	5	6	3

S2-15

5	2	1	3	4	6
4	6	3	1	5	2
6	1	5	4	2	3
2	3	4	5	6	1
3	5	6	2	1	4
1	4	2	6	3	5

S2-16

1	4	2	6	3	5
6	3	5	1	4	2
5	1	4	2	6	3
2	6	3	5	1	4
3	5	1	4	2	6
4	2	6	3	5	1

S2-16

2	4	5	1	6	3
6	3	1	5	2	4
5	6	3	4	1	2
1	2	4	3	5	6
4	5	6	2	3	1
3	1	2	6	4	5

S2-17

3	2	6	5	1	4
5	4	1	3	6	2
6	3	4	1	2	5
1	5	2	6	4	3
4	6	5	2	3	1
2	1	3	4	5	6

S2-18

3	2	1	6	5	4
4	6	5	2	1	3
6	1	4	5	3	2
2	5	3	1	4	6
1	3	6	4	2	5
5	4	2	3	6	1

S2-18

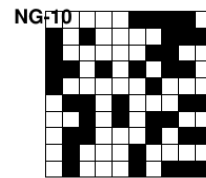
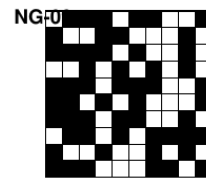
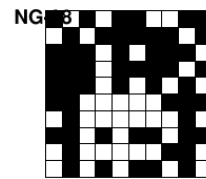
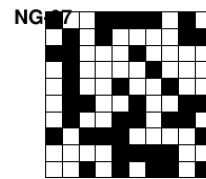
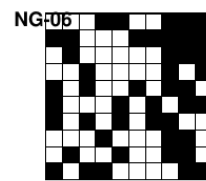
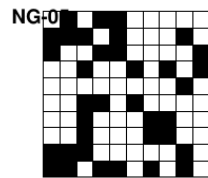
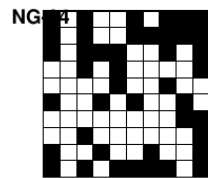
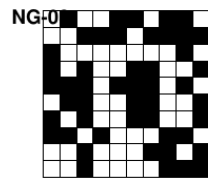
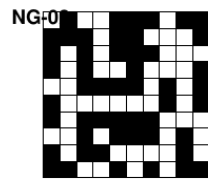
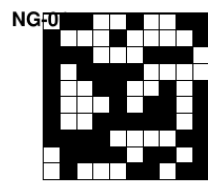
5	2	4	1	3	6
5	1	3	2	4	6
4	6	1	5	2	3
3	5	2	6	1	4
2	3	6	4	5	1
1	4	5	3	6	2

S2-20

5	3	2	4	1	6
4	1	6	5	3	2
6	4	3	2	5	1
2	5	1	6	4	3
3	6	5	1	2	4
1	2	4	3	6	5

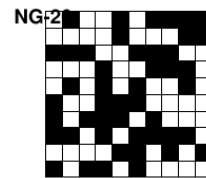
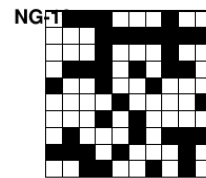
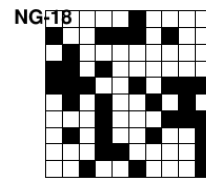
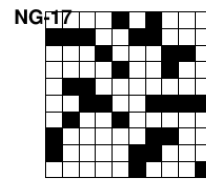
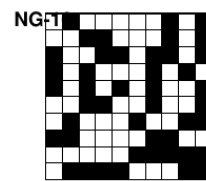
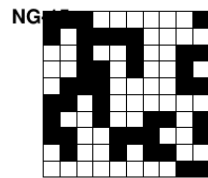
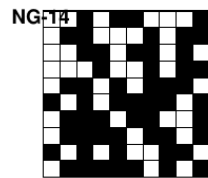
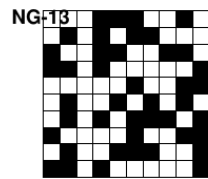
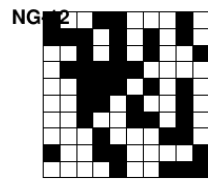
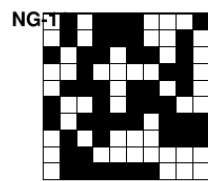
SOLUTIONS

Solutions Page 3/10



SOLUTIONS

Solutions Page 4/10



SOLUTIONS

Solutions Page 5/10

BP-01

0	1	0	0	0	1	1	1	0	1	0
1	1	1	0	0	0	0	1	0	1	1
0	1	1	0	0	0	0	1	1	1	1
0	1	0	1	1	0	1	1	0	1	0
1	0	1	1	1	0	0	0	0	1	1
0	0	0	1	1	1	1	0	1	0	1
1	1	0	1	1	1	1	0	0	0	0
0	0	1	1	1	0	0	1	1	0	1
1	1	1	0	0	1	0	1	0	1	0
0	0	0	0	1	1	0	1	1	1	1

BP-02

0	0	1	1	1	1	0	0	0	1	0
0	0	1	1	0	1	0	1	1	1	0
1	1	1	0	1	0	0	0	0	1	1
0	0	1	1	1	0	0	0	1	1	1
1	0	1	0	0	1	1	0	0	1	1
0	1	0	0	0	1	1	1	1	1	0
1	1	0	1	1	0	0	0	1	1	0
0	1	0	0	1	0	1	1	0	1	1
1	0	0	1	0	1	1	1	0	0	1
0	1	0	0	0	1	1	1	0	1	1

BP-03

0	1	1	0	1	1	0	0	0	1	1
0	1	1	0	1	0	1	0	1	1	0
1	1	0	0	0	1	0	1	1	1	0
0	0	0	1	1	1	0	0	1	1	1
1	0	1	1	0	0	0	1	0	1	1
0	1	0	1	0	0	1	1	1	1	0
0	1	0	0	0	1	1	1	1	1	0
1	0	1	1	1	0	0	0	0	1	1
1	0	1	0	1	0	0	0	1	1	1
1	0	0	1	0	1	1	1	0	0	1

BP-04

0	0	0	0	1	0	1	0	1	1	1
0	1	1	1	1	0	0	0	0	1	1
0	1	1	1	1	0	0	1	0	0	0
0	0	0	0	0	1	1	1	1	1	1
1	1	1	1	0	1	0	0	0	0	0
1	0	0	0	1	1	0	1	1	0	1
1	0	0	0	0	1	1	1	1	0	1
0	1	1	0	1	0	0	0	1	1	1
1	1	0	1	0	0	1	0	0	1	1
0	0	1	1	0	1	1	1	0	0	1

BP-05

0	1	1	0	1	1	0	0	0	0	0
0	1	0	1	1	1	1	0	0	0	0
0	1	1	1	1	0	0	1	0	0	1
0	1	1	0	0	1	0	1	1	1	0
0	1	0	0	1	0	1	1	0	1	1
0	1	1	1	0	1	1	0	0	0	0
1	0	0	1	1	0	0	0	1	1	1
1	0	0	0	0	0	1	1	1	1	1
1	0	1	0	0	1	0	0	1	1	1
1	0	0	0	1	0	0	1	1	1	1

BP-06

0	0	1	1	0	0	0	1	0	1	1
1	0	1	0	1	0	1	1	1	0	0
1	1	0	0	1	1	1	0	0	0	0
0	1	0	0	1	1	0	1	1	1	0
0	0	1	1	1	0	1	0	0	0	1
1	0	1	1	0	0	1	0	0	1	1
1	1	0	1	0	1	0	0	1	0	1
0	1	0	0	0	1	0	1	1	1	1
0	0	1	0	1	1	0	1	1	1	0
0	1	0	1	0	0	1	0	1	1	1

BP-07

0	0	0	0	1	0	1	0	1	1	1
1	1	1	1	0	0	0	1	0	0	0
1	1	0	1	0	1	0	0	1	0	0
0	1	1	0	1	1	0	1	0	0	0
0	0	0	0	1	1	1	1	1	0	0
0	0	1	0	1	1	1	1	0	0	0
1	1	0	1	1	0	0	0	0	0	1
0	0	1	1	0	0	1	1	0	1	0
0	0	1	0	0	1	1	0	1	1	1
1	1	0	1	0	0	0	0	1	1	1

BP-08

0	0	0	1	0	1	1	1	1	0	0
1	0	1	1	1	0	1	0	0	0	0
0	1	1	0	0	0	1	0	1	1	1
0	1	0	1	0	1	0	1	1	1	0
0	0	0	1	1	1	1	1	0	0	0
1	0	1	1	0	0	0	0	1	1	1
1	0	0	0	1	1	0	1	0	1	1
0	1	0	1	0	0	1	1	1	0	0
1	0	1	0	1	1	0	0	0	1	1
1	1	1	0	0	1	0	0	0	1	1

BP-09

0	0	0	0	1	1	1	1	0	0	0
0	0	1	1	0	1	0	1	0	1	1
0	0	1	1	0	0	0	1	1	1	1
1	1	0	0	1	1	1	0	0	0	0
0	1	0	1	1	0	1	0	1	0	0
1	0	1	0	0	0	1	1	0	1	1
0	0	0	1	1	1	0	0	1	1	1
1	0	1	1	0	1	0	0	0	0	0
0	1	1	0	0	0	0	1	1	1	1
1	1	0	0	1	1	0	0	1	0	0

BP-10

0	1	1	1	1	0	0	0	0	0	0
1	1	1	0	1	0	1	0	0	0	0
0	0	0	0	0	1	1	1	1	1	1
1	1	0	1	0	0	1	1	0	0	0
1	1	0	0	0	0	1	1	1	0	0
0	0	1	1	1	1	0	0	0	1	1
1	0	1	0	1	0	0	0	1	1	1
0	0	1	1	0	1	0	1	0	1	1
1	1	0	0	0	1	1	0	1	0	0
0	0	0	1	1	0	0	1	1	1	1

SOLUTIONS

Solutions Page 6/10

BP11

1	0	1	0	1	1	0	0	1	0
0	1	1	0	0	1	0	1	1	0
0	1	1	1	1	0	1	0	0	0
1	0	0	0	0	1	1	1	0	1
1	0	0	1	0	1	0	0	1	1
0	0	0	1	0	1	1	1	0	1
1	1	0	1	1	0	0	0	1	0
0	1	1	0	0	0	1	1	0	1
0	1	1	1	1	0	0	0	1	0
1	0	0	0	1	0	1	1	0	1

BP12

1	0	1	1	0	1	0	1	0	0
1	0	1	0	1	0	0	1	1	0
0	0	0	1	1	1	0	1	1	0
1	1	0	1	0	1	0	0	0	1
1	0	0	1	1	1	0	1	0	0
0	1	0	0	0	1	1	0	1	1
0	1	1	0	0	0	1	0	1	1
0	1	1	0	0	0	1	1	1	0
1	0	1	0	1	0	1	0	0	1
0	1	0	1	1	0	1	0	0	1

BP13

1	0	1	1	0	1	0	1	0	0
0	1	0	1	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0	1
0	1	0	0	1	1	1	0	1	0
0	1	1	1	1	0	0	1	0	0
1	0	0	0	1	0	1	0	1	1
0	1	0	0	1	0	1	0	1	1
1	0	0	1	0	1	0	1	0	1
1	0	1	0	0	1	1	1	0	0
0	0	1	0	1	0	1	1	1	0

BP14

1	1	1	1	1	0	1	0	0	0
1	1	1	0	0	1	0	1	0	0
1	0	0	0	0	1	1	1	0	1
1	1	0	0	1	0	1	0	0	1
1	0	0	1	0	1	0	1	1	0
1	1	0	0	0	1	1	1	0	0
0	0	1	1	1	0	0	0	1	1
0	0	0	1	1	1	1	0	1	0
0	0	1	1	0	0	0	1	1	1
0	1	1	0	1	0	0	0	1	1

BP15

1	0	0	1	1	1	1	0	0	0
0	0	1	1	0	1	0	1	1	0
0	0	1	0	1	0	1	1	1	0
1	1	1	0	1	0	1	0	0	0
1	0	0	1	0	1	0	0	1	1
1	1	0	1	0	0	1	0	0	1
1	1	1	1	0	0	0	0	0	1
0	0	1	0	1	1	0	1	1	0
0	1	0	0	1	1	0	0	1	1
1	0	0	1	0	0	1	1	0	1

BP16

1	1	1	1	0	0	1	0	1	1
1	1	1	0	0	0	1	1	0	0
0	0	1	1	0	0	1	1	0	1
1	1	1	0	0	1	0	0	1	0
1	1	0	0	0	1	1	0	1	0
1	0	0	0	1	0	1	1	0	1
1	0	0	1	1	0	0	1	1	0
0	1	0	1	1	1	0	0	1	0
0	1	0	0	0	1	1	0	1	1
0	0	1	1	1	1	0	0	0	1

BP17

1	1	1	1	0	0	1	0	0	0
0	0	0	1	1	0	1	0	1	1
0	0	1	1	0	0	1	1	0	1
1	1	0	1	0	1	1	0	0	0
1	1	0	0	1	1	0	1	0	0
1	1	1	0	0	1	0	0	1	0
0	0	1	1	1	0	0	1	0	1
1	1	0	0	0	1	1	0	1	0
0	0	0	0	1	1	0	1	1	1
1	0	1	0	0	0	1	0	1	1

BP18

0	1	1	1	1	0	0	0	0	0
1	0	1	1	0	1	0	0	1	0
0	1	0	1	0	1	0	1	1	0
1	1	0	0	1	0	0	1	0	1
0	0	1	0	0	0	1	1	1	1
1	1	1	0	0	0	0	1	1	0
1	0	0	0	1	1	1	0	0	1
0	0	1	0	1	0	1	1	0	1
1	1	0	1	0	1	0	0	1	0
0	0	1	1	1	0	1	0	0	1

BP19

0	1	0	1	1	1	1	1	0	0
1	0	0	1	1	0	1	0	0	1
0	1	1	1	0	1	0	0	0	1
0	1	0	1	0	1	0	1	1	0
1	1	0	0	1	0	0	1	0	1
1	1	0	0	0	1	0	1	1	0
1	0	1	0	1	0	1	0	1	0
0	0	1	1	0	0	1	0	1	1
1	0	1	0	1	0	1	0	0	1
0	1	1	0	1	1	0	1	0	0

BP20

0	1	0	1	0	1	0	0	0	1
0	0	1	0	1	1	1	1	0	0
1	0	0	1	1	0	0	1	1	0
0	0	1	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0	0	1
1	0	0	0	1	0	1	1	1	0
1	1	0	0	0	0	1	1	1	0
1	0	1	1	1	0	1	0	0	0
0	1	0	0	0	1	1	1	0	1
0	1	1	0	0	1	0	0	1	1

SOLUTIONS

Solutions Page 7/10

FT-08

8	1	2	4	5	3
3	4	6	5	1	2
5	6	4	3	2	1
1	3	5	2	6	4
4	2	1	6	3	5
2	5	3	1	4	6

FT-08

2	3	5	4	6	1
1	2	3	5	4	6
3	5	4	6	1	2
5	4	6	1	2	3
4	6	1	2	3	5
6	1	2	3	5	4

FT-08

1	5	4	2	6	3
4	3	2	6	5	1
3	6	1	4	2	5
6	4	5	3	1	2
2	1	6	5	3	4
5	2	3	1	4	6

FT-08

2	4	6	3	1	5
1	3	2	5	6	4
6	5	1	4	2	3
3	2	4	1	5	6
5	1	3	6	4	2
4	6	5	2	3	1

FT-08

5	1	3	6	2
1	3	4	5	2
2	1	6	4	3
5	6	3	2	4
6	4	2	1	5
3	2	5	6	1

FT-08

8	2	5	4	1	3
3	1	6	2	5	4
2	6	4	5	3	1
5	4	1	3	2	6
1	3	2	6	4	5
4	5	3	1	6	2

FT-08

3	1	4	6	2	5
6	2	3	5	4	1
4	5	2	3	1	6
1	3	5	2	6	4
2	6	1	4	5	3
5	4	6	1	3	2

FT-08

2	3	5	1	6	4
6	2	4	5	3	1
1	4	6	2	5	3
3	6	1	4	2	5
5	1	3	6	4	2
4	5	2	3	1	6

FT-08

2	5	4	1	6	3
5	2	1	4	3	6
1	4	6	3	5	2
3	6	5	2	4	1
4	1	3	6	2	5
6	3	2	5	1	4

FT-08

5	2	4	3	6	1
4	3	2	1	5	6
2	1	3	6	4	5
6	4	5	2	1	3
3	6	1	5	2	4
1	5	6	4	3	2

SOLUTIONS

Solutions Page 8/10

FT-10

2	1	5	3	6	4
1	3	6	2	4	5
4	5	1	6	3	2
6	4	2	5	1	3
5	6	3	4	2	1
3	2	4	1	5	6

FT-12

5	4	6	1	2	3
2	5	1	3	4	6
4	2	3	6	5	1
3	1	5	2	6	4
1	6	4	5	3	2
6	3	2	4	1	5

FT-14

4	5	6	1	2	3
2	3	4	5	6	1
5	2	1	4	3	6
1	4	3	6	5	2
6	1	2	3	4	5
3	6	5	2	1	4

FT-18

5	1	2	3	4	6
3	6	1	4	5	2
2	4	3	1	6	5
1	5	4	6	2	3
4	2	6	5	3	1
6	3	5	2	1	4

FT-19

2	1	3	4	6	5
4	3	6	5	1	2
3	4	5	6	2	1
5	6	1	2	3	4
6	5	2	1	4	3
1	2	4	3	5	6

FT-16

2	5	6	1	3	4
3	2	4	5	6	1
5	1	3	4	2	6
1	4	2	6	5	3
6	3	1	2	4	5
4	6	5	3	1	2

FT-17

6	2	1	5	3	4
2	1	6	3	4	5
3	4	5	1	6	2
5	3	4	2	1	6
4	5	3	6	2	1
1	6	2	4	5	3

FT-13

3	2	6	5	4	1
5	3	1	2	6	4
1	6	2	4	5	3
6	4	5	1	3	2
4	1	3	6	2	5
2	5	4	3	1	6

FT-15

5	2	3	4	6	1
1	6	5	2	4	3
4	1	6	5	3	2
3	4	1	6	2	5
2	3	4	1	5	6
6	5	2	3	1	4

FT-28

5	6	4	2	1	3
6	4	5	3	2	1
3	1	2	6	5	4
1	2	3	4	6	5
2	3	1	5	4	6
4	5	6	1	3	2

SOLUTIONS

Solutions Page 9/10

N01 Next = 3072

N02 Next = 13

N03 Next = 32

N04 Next = 31

N05 Next = 262144

N06 Next = 17

N07 Next = 972

N08 Next = 22

N09 Next = 27

N10 Next = 33

SOLUTIONS

Solutions Page 10/10

N11 Next = 117649

N12 Next = 19

N13 Next = 32

N14 Next = 11

N15 Next = 43

N16 Next = 32

N17 Next = 64

N18 Next = 37

N19 Next = 1024

N20 Next = 31